

Impacts of Climate Change on the Widespread and Epidemics of Some Tomato Diseases during the Last Decade in Egypt

M.A. Fahim and M.K. Hassanein
Central Laboratory for Agricultural Climate
PO Box 296 Dokki, 12411 Giza
Egypt

A.F. Abou Hadid and M.S. Kadah
Climate Change Information Center
9 El-Gamaa st., 12619 Giza
Egypt

Keywords: extreme weather, climate trends, disease epidemic, positive impacts

Abstract

This study explores some of the impacts that climate change, with warmer and more variable weather, may have on tomato diseases in Egypt. Its aim is to elucidate how warmer temperatures and an increase in extreme weather events may affect the incidence of some tomato fungal diseases in recent years. Recent climate trends, such as increased nighttime and winter temperatures, may be contributing to the greater prevalence of tomato diseases. The ranges of several important tomato disease in Egypt; including tomato late blight (the most destructive tomato disease causing fruit yield losses) have expanded since the early 1990s, possibly in response, in part, to climate trends. Wet vegetation promotes the germination of spores and the proliferation of fungi. Based on analysis of plant/disease/climate relations, an epidemic of late blight onset on tomatoes that is 1-2 weeks earlier means 2-3 additional sprays to achieve sufficient control of late blight. Accordingly, 1-3 more sprays will be applied at the incoming decades of the 2025-2100. The combination of long-term change (warmer average temperatures) and greater extremes suggest that climate change could have positive impacts on main tomato diseases in Egypt. There is a need for a better impact assessment of climate change on plant pests and diseases. In the Fourth Assessment Report of the IPCC4 these risks are insufficiently addressed.

INTRODUCTION

The risks associated with agriculture and climate change is a result of the strong complicated relationships between agriculture and climate system, plus the high reliance of the agriculture system on natural resources. The ranges of several important crop pests in Egypt; including the potato and tomato late blight (the most destructive disease causing yield losses) have expanded since the early 1990s, possibly in response, in part, to climate trends (Fahim et al., 2007). Recent climate trends, such as increased nighttime and winter temperatures, may be contributing to the greater prevalence of crop pests (Fahim, 2010). Higher temperatures in some regions are likely to result in the spread of plant pests and diseases. Higher temperatures reduce insect winterkill, and lead to increased rates of development and shorter times between generations (Coakley et al., 1999). Wet vegetation promotes the germination of spores and the proliferation of bacteria, fungi, and nematodes. There is a need for a better impact assessment of climate change on plant pests and diseases. In the Fourth Assessment Report of the IPCC these risks are insufficiently addressed (IPPC 2007).

This study explores some of the impacts that climate change, with warmer and more variable weather, may have on tomato late blight disease in Egypt as one of the polycyclic epidemic diseases.

MATERIALS AND METHODS

Tomato Late Blight Assessment

As per the data observed by Fahim et al. (2007), the assessed tomato cultivars were 'Castel rock', 'Super strain B', 'UC97' and 'B2\86'. Crops were assessed for foliar, stem and fruit blight in the field in winter and early summer. First assessment within

60-70 days after planting (DAP) and usually in December and January for the winter season, and during April and May for the early summer season.

Climate Change Projection Data

Projection climate data for the period of 2025 to 2100 are a result of the simple climate model simulations representing data for future time series (e.g., 2025s, 2050s, 2075s and 2100s). Climate change scenarios were assessed according to future conditions derived from MAGICC/SCENGEN software of the University of East Anglia (UK). The data are used to predict the late blight disease on tomato throughout the coming decades from 2025 to 2100 in Egypt. The method of Hannukkala et al. (2007) was used to illustrate the trend mean of the changed temperature.

Growing Day-Degree (GDD) Calculation

Growing day-degree (GDD) was calculated by Kaukoranta (1996) method. Days after planting (DAP) were used as the starting point for calculation of day-degree (GDD) with a base temperature of 10°C. GDD was calculated and applied to project if climate change will effect on the changes in the onset of epidemics in the incoming periods 2025, 2050, 2075 and 2100. The Accu.GDD was calculated from projected daily maximum and minimum temperature by A1, B1, A2 and B2 scenarios for the above mentioned periods.

Effect of Climate Change on the Onset of Epidemics

The estimated regression parameters applied to project climate change will have an effect on the onset of epidemics in the incoming periods from 2025 to 2100. The method of Kaukoranta (1996) was used to develop degree-day models for determining the percent of the late blight outbreak versus the current level of the disease. The method suggested that the potential yield gain was almost completely offset when late blight was considered.

RESULTS AND DISCUSSION

Tomato Late Blight Observation in Egypt

The late blight on tomato was detected earlier in the winter and early summer seasons of 2006 than 2005 by 2-4 weeks (Fig. 1a). In terms of odds ratios, the risk of disease in the seasons of 2006 (epidemic seasons) was 2.74 and 7.15-fold greater than for the seasons of 2005 (non-epidemic seasons). For all seasons studied, the risk of the late blight was 2.60-fold greater in the eastern Delta (e.g., Salhia) compared with the southern parts of the Delta (e.g., Badrashin). The disease progress distribution seems moved from south to west center to east of the Delta (Fig. 1b).

Changing in Tomato Late Blight Due to Climate Change

Figure 2a,b shows the accumulative day-degree required for first appearance of tomato late blight. Accu.GDD were 72 and 212, respectively, for Badrashin area during the early summer season. However, in Salhia during the winter season they were 496 and 637 Accu.GDD. Formal tests have been used to evaluate statistical assumptions. The coefficient of determination R^2 ranged from 0.953-0.985. Adjusted R^2 have been compared and the models had slightly higher value (Fig. 3).

The Accu.GDD model described in this study has proven effective and popular for timing initial control and the Accu.GDD threshold considered empirically selected reflected a compromise among a number of variables such as planting date. The time of first lesion appearance of *Phytophthora infestans* is closely related to plant maturity.

The increase in tomato late blight to a mere 6% of its current level of the epidemic is due to the projection of climate change. Late blight outbreaks projected to occur from 10 to 15 days earlier in terms of crop susceptible was lengthened by 10 to 20 days during warming conditions (Fig. 4a,b).

Identification of Gaps Related to Climate Change Impacts on Pests and Diseases

To understand the contribution of climate change to crop pest outbreaks, there is a need to: i) establish benchmarks of current biosecurity status and impacts and costs of biosecurity; and ii) monitor indicators of change in biosecurity in relation to rates of invasions by foreign species, rates of crop, livestock, forest and fish losses, and changes in costs of biosecurity. However, the costs associated with this may be prohibitive.

CONCLUSIONS

The potential impact of climate change on the development and growth of crops and pests was investigated by means of computer modeling. However, little research has been conducted into the effects of climate change on plant diseases compared with other studies. Meanwhile, the obtained results of the current study created new opportunities to study such effects by methodological development. Meantime, the earlier onset of warm temperatures could result in an earlier threat from disease with the potential for more severe epidemics and increases in the number of fungicide applications needed for control.

ACKNOWLEDGEMENTS

Thanks to the staff of the Central Laboratory for Agricultural Climate and the Climate Change Information Center, Cairo, Egypt for technical and positive advice.

Literature Cited

- Coakley, S.M., Scherm, H. and Chakraborty, S. 1999. Climate change and disease management. *Ann. Rev. Phyto.* 37:399-426.
- Fahim, M.A. 2010. The impact of climate change on crop pests and diseases in Egypt. National consultant report of UNJP/EGY/022 project, FAO.
- Fahim, M.A., Medany, M.A., Aly, H.A. and Fahim, M.M. 2007. Effect of the climate change on the widespread and epidemics of potato and tomato late blight disease under the Egyptian conditions. Ph.D. Thesis, Cairo Univ., Egypt, p.177.
- Hannukkala, A.O., Kaukoranta, T. and Rahkonen, A. 2007. Late-blight epidemics on potato in Finland, 1933-2002; increased and earlier occurrence of epidemics associated with climate change. *Plant Pathol.* 56:167-176.
- IPCC. 2007. *Climate Change 2007: Impacts, Adaptation and Vulnerability*. the FAR of the IPCC, Cambridge University Press, Cambridge, UK, p.1000.
- Kaukoranta, T. 1996. Impact of global warming on potato late blight: risk, yield loss and control. *Agric. Food Sci. Finl.* 5:311-27.

Figures

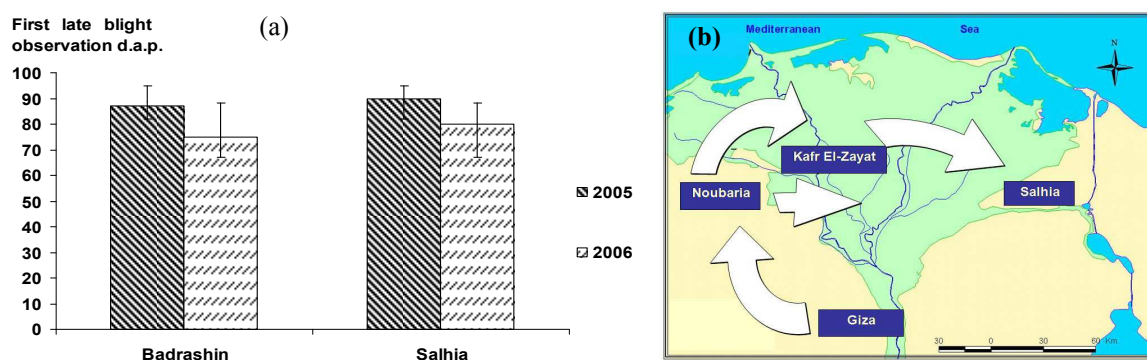


Fig. 1. Mean number of days after planting to first disease observation at observation locations (a). Tomato late blight progress distribution (b).

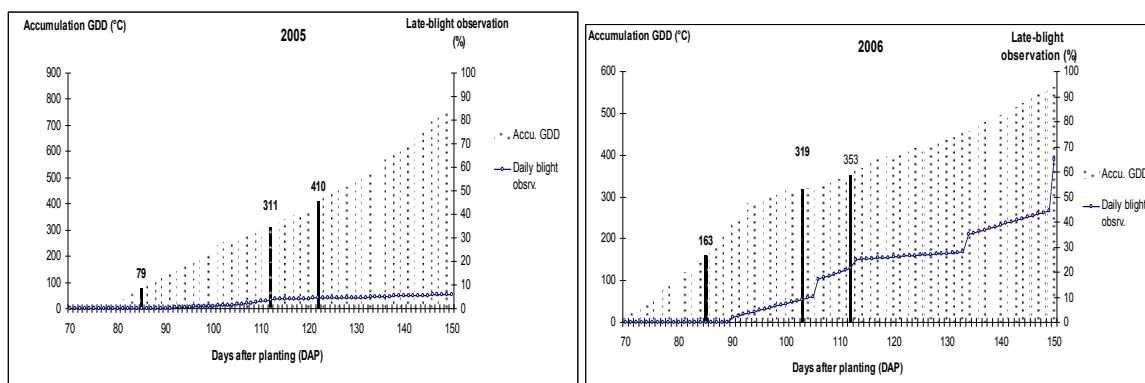


Fig. 2. Growing degree-day (Accu.GDD) relative to disease observation for determining first appearance of tomato late blight for the seasons of 2005 and 2006. Dark columns indicate to the three stages of the epidemic.

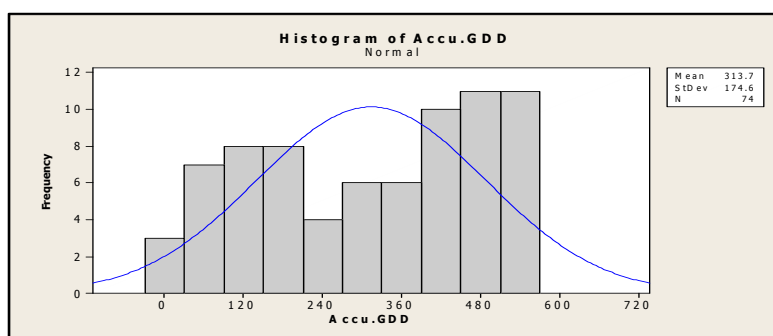


Fig. 3. Normal distribution of the statistical analysis of Accu.GDD by Minitab Inc. © 2006.

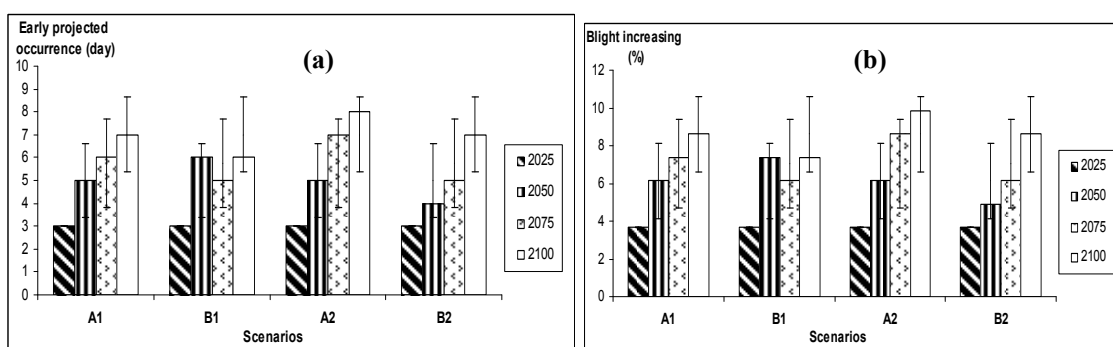


Fig. 4. Suggested increase in the disease level to the current level of the tomato late blight epidemic in relation to different illustrative SRES climate change scenarios (e.g., A1, A2, B1 and B2). (a) is early projected occurrence days. (b) is the % projected increase of late blight.